Vipul Sapra

Kharghar, Navi Mumbai | 8888-033-168 | vpsapra5@gmail.com | linkedin.com/in/vipul-sapra-ab30a09a

EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Dimensional Data Modeling, Star/Snowflake Schemas, SCD Type 1/2, Regulatory Compliance Analytics, Predictive Modeling, Anomaly Detection, Generative AI & LLM Automation, Data Governance, ETL/ELT Pipeline Architecture, Statistical KPI Calibration, Executive Reporting & KOM Dashboards.

Programming Languages & Tools: Azure Data Factory (ADF), Azure Synapse, Azure SQL, Databricks, AWS S3, AWS Redshift, Snowflake, Apache Spark, PySpark, Python, Scikit-Learn, XGBoost, SARIMA, Power BI, Tableau, MicroStrategy, Airflow, One Automation, SQL.

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Lead Business Intelligence & Data Architecture Engineer — **May 2022 – Present**

- **Architected end-to-end scalable BI solutions** utilizing high-performance SQL stored procedures, dimensional data modeling, and Star/Snowflake schemas to power complex regulatory compliance, cost-optimization, and operational telemetry across multi-terabyte datasets.
- **Engineered automated data quality audit frameworks** with real-time firewalls and proactive anomaly detection machine learning models achieving 95% accuracy, mitigating ETL discrepancies and reducing operational turnaround times by over 80%.
- **Developed advanced predictive and statistical models** including XGBoost, Random Forests, and SARIMA forecasting to monitor metrics, optimize workforce capacity planning, and drive data-backed decision support across global business units.
- **Spearheaded GenAI and agentic AI-assisted development** leveraging prompt engineering and intelligent regex classification to automate regulatory data extraction, profiling, and compliance reporting workflows, accelerating feature delivery by 40%.
- **Delivered executive-facing KOM dashboards and analytics** using Power BI, Tableau, and MicroStrategy, translating complex multi-jurisdictional telemetry into actionable strategic insights while maintaining 99.9% data availability SLAs.

Master of Science in Information Systems & Data Science Graduate — **May 2022 – Present**

- **Mastered advanced statistical modeling and data warehousing** through rigorous coursework at the University of Cincinnati, achieving a 3.95/4.0 GPA with specialized focus on information retrieval, machine learning, and scalable architecture.
- **Applied advanced machine learning algorithms** to complex datasets, mastering Scikit-Learn, predictive regression, and data mining techniques for automated anomaly detection and customer behavior modeling.
- **Executed comprehensive data governance and analytics projects** bridging business stakeholder requirements with high-throughput database systems and modern visualization platforms.

Bachelor of Engineering in Petrochemical — **May 2022 – Present**

- **Built foundational quantitative and analytical problem-solving** skills at Pune University MIT through rigorous engineering mathematics, statistical analysis, and complex process optimization.
- **Modeled multi-variable physical and operational systems** requiring precision data collection, rigorous error analysis, and cross-functional root cause investigations.
- **Optimized resource allocation and workflow efficiency** by translating raw operational metrics into structured engineering frameworks and actionable reporting.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.